

Symposium: Eye/orbital brachytherapy: from organ sparing to function sparing

SP-0034 Brachytherapy Choice of the best approach for eye melanoma

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Abstract Text

Melanoma of the eye can be intraocular (uveal), ocular surface, in the orbit or on the eyelid. There is a range of possible treatments for uveal melanoma. These include laser (thermotherapy or photodynamic therapy), radiotherapy (brachytherapy, proton beam radiotherapy, stereotactic radiosurgery) and surgery (tumour resection or enucleation).

Plaque brachytherapy is designed to deliver radiotherapy to uveal melanoma in a controlled way. The aim is to treat the tumour, and to minimise risks of vision or eye loss. The Collaborative Ocular Melanoma Study (COMS) described the standard for brachytherapy using 125-I, defined the size of melanomas and importantly showed equivalent survival in a randomised controlled trial between brachytherapy versus plaque radiotherapy for medium sized melanomas.

The American Brachytherapy Society's consensus guidelines for the treatment of intraocular tumours suggest certain parameters. For uveal melanoma the dose prescribed is 70 to 100 Gy to the tumour apex. The prescription isodose line should encompass the entire tumour. Dose rates should not be less than the COMS historical standard of 0.60 Gy/hour. Isotope choice is limited by the size of tumour, with 106-Ru plaques for tumours less than 6mm height. 125-I or 103-Pd can treat larger tumours, but tumours greater than 12 mm in apical height or 20 mm in base carry guarded prognosis for retaining useful vision. Alternatives to brachytherapy for unsuitable cases where the plaque cannot be placed easily or the tumour is too large, include proton beam radiotherapy, stereotactic radiosurgery or enucleation of the eye. Failure of brachytherapy can be managed by further radiotherapy (brachytherapy or proton beam), laser (usually thermotherapy) or enucleation of the eye.

Plaque radiotherapy is an established technique for the treatment of uveal melanoma. Many eyes that would otherwise be lost are treated with this method. The challenge remains to minimise complications and to save more lives.

SP-0035 The role of interstitial radiotherapy in orbital tumors or extrascleral uveal melanoma

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Abstract Text

introduction

Radiation therapy continues to play an essential role in the management of benign and malignant orbital tumors of children and adults.

Material and methods

Multidisciplinary perioperative and interstitial interventional radiotherapy (IRT) after extensive but visual acuity preserving tumor resection is reported as safe and well tolerated treatment of advanced/recurrent malignancies involving the intraorbital space and of the skull base. If comparing modern EBRT and IRT technologies, the advantage of IRT was stated also in the adjuvant setting.

Results

Interstitial IRT can be applied as monotherapy or complementary to external beam radiation as a local dose escalation at the external radiation field margins or functionally important sites with micro-, and/or macroscopic residual tumor masses. Nowadays, stepping source techniques with personalized dose planning dominate the field. Visual acuity preservation become possible due to multidisciplinary teamwork in IRT excellence centers of specialized orbita groups. Especially in children intraorbital tumors, surgery combined with intraorbital interventional radiotherapy was proven by several groups as an optimal visual acuity preservation treatment method.

Furthermore, regardless of the invasive character of the treatment, it can be used as adjuvant local treatment following enucleation in cases of extraocular infiltrating melanomas, since the toxicity profile is more advantageous compared to any other modern external beam technique. Additionally, by avoiding exenteration, better cosmetic rehabilitation becomes possible. Although, adjuvant postenucleation EBRT with modern technology has low toxicity rates, review works in the literature advise the use of IRT because of improvements in toxicity results.

Conclusion

Interstitial, multidisciplinary IRT plays an important role in visual acuity preservation treatments of intraorbital tumors and offers the lowest toxicity profile in the adjuvant radiotherapy of enucleated extrascleral uveal melanomas.

SP-0036 Brachytherapy toxicity: diagnosis, therapies and preventive strategies

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Abstract Text

Purpose: to illustrate diagnostic methods, therapies and preventive strategies for radiation-induced side effects, after brachytherapy. The benefits of saving the eye with brachytherapy may be reduced by visual function impairment, secondary